

dwg: detect related pairs on the GRM, then whiten and centre

genotypes G

standardise,
not centred

GRM A
(scaled, not centred)

fit
 $A = L + S$

L : diag. + pairs imputed
by top 3 PCs (rank 4)

S : diagonal + pairs
 $\hat{\phi} > \tau$ (scaled by v)

+

noise v = diagonal of S ; families from the pairs

Σ : noise v
+ families

whiten

whitened $\Sigma^{-1/2} A \Sigma^{-1/2}$
(not centred: all positive)

centre

centred: $\Sigma^{-1/2} \mathbf{1}$
projected out

top k ,
 $\times \Sigma^{1/2}$

PCs: PC2 vs PC3
MXL axis $R^2 = 1.00$

PC3

PC2

● CEU ● CHB ● MXL ● YRI ★ relative